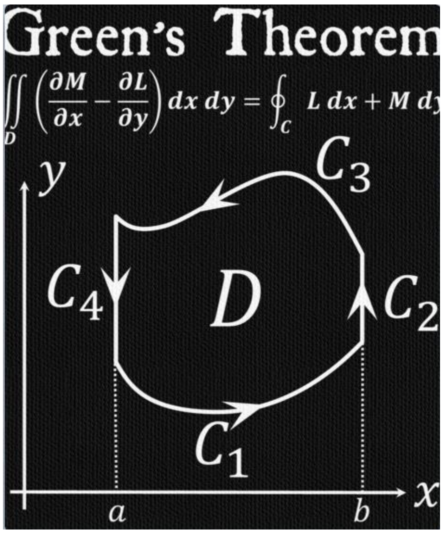
REDHILL SCHOOL

Tour de Maths 2024

Leg 1

Solutions



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AMT.

PRESH TALWALKAR

5 MARK QUESTIONS

QUESTION 1

A father's age is currently three times that of his daughter. In twelve years, he will be twice as old as his daughter will be at that time. What is the current age of the father?

	Age now	*Age in 12 years*
Father	$3x$	$3x+12$
Daughter	x	$x+12$

$3x+12=2(x+12)$

$\therefore x=12$

Daughter is 12 and father is 36.

Answer: *36*

QUESTION 2

How many numbers from 1 to 1000 are divisible by 3 or 5?

(A) 467 (B) 500 (C) 533 (D) 600

$$\begin{aligned} n(E) &= n(\div 3)+n(\div 5)-n(\div 15) \\ &= \frac{1000}{3}+\frac{1000}{5}-\frac{1000}{15} \\ &= 467 \end{aligned}$$

Answer: *A*

QUESTION 3

At a party there were 110 people who were either female or male. Each person could dress in either blue or green.

In addition, the following information has been given:

- 35 males are dressed in green.
- 50 are females.
- 45 are dressed in blue.

How many females are dressed in green?

$110 - 35 - 45 = 30$

Answer: *30*

QUESTION 4

What is the highest number below 100 000, which when written in words, doesn't have the letter ***n*** in it?

Every number over one hundred has the word 'hundred' in them, which contains the letter 'n', same with 'thousand'. The nighties also have the letter 'n' because of 'ninety'. The highest number without the letter n is 'eighty-eight'.

Answer: *88*

QUESTION 5

If it took 135 minutes for 5 t-shirts to dry on a washing line, how long would it have taken for 17 t-shirts to dry?

The t-shirts are the same on the same washing line so it doesn't matter how many they are, logically they should take the same amount of time as it took the 5 t-shirts.

Answer: *135 Minutes*

QUESTION 6

Evaluate:

$$2024^{1^{2^{3^{4^{5}}}}}$$

1 raised to the power of any exponent is 1.

$1^{2^{3^{4^{5}}}} = 1$

$\therefore 2024^{1^{2^{3^{4^{5}}}}} = 2024^{1}$

Answer: *2024*

QUESTION 7

A “lucky” number between 1 and 50 meets the following criteria:

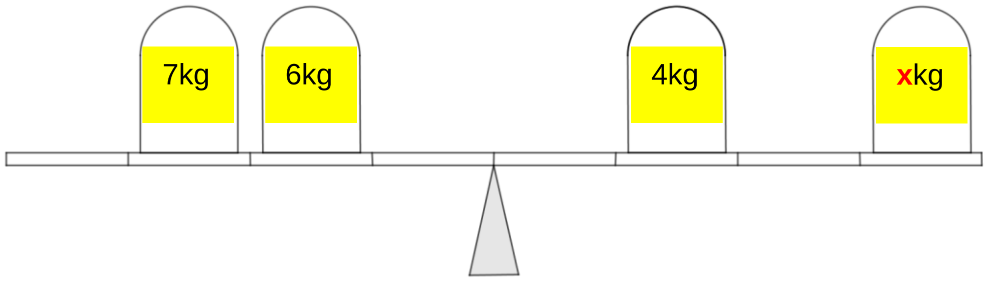
- It is divisible by 3 when the digits are added together.
- the total is between 4 and 8.
- It is an odd Number.
- When the digits are multiplied together the total is between 4 and 8.

What is the lucky number?

(A) 12 (B) 18 (C) 27 (D) 15

Answer: *D*

QUESTION 8



What mass should be placed on **x** to balance the scale?

Left arm:

$7kg \times 3 = 21$

$6kg \times 2 = 12$

$Total = 33$

Right arm:

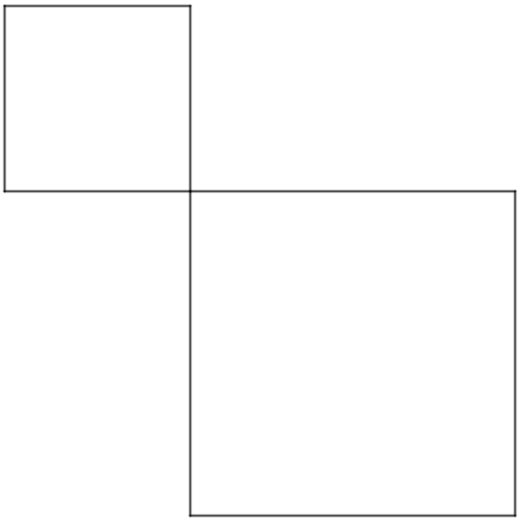
$4kg \times 2 = 8$

$6.25kg \times 4 = 25$

$Total = 33$

Answer: *6,25kg*

QUESTION 9



An area of land, consisting of the sums of the two squares, is 1000 square metres. The side of one square is 10 metres less than two-thirds of the side of the other square. What are the sides of the two squares?

$1000 = 10^2 + 30^2$

Answer: *10 and 30 metres.*

QUESTION 10

A maths teacher writes the following on the whiteboard:

What will the units digit be of this number?

$4^1 = 4$; $4^2 = 16$; $4^3 = 64$; $4^4 = 256$; ...

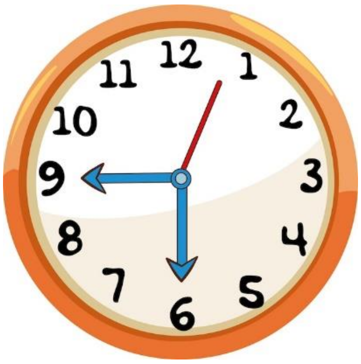
The unit digit of 4^x, $x \in \mathbb{Z}^+$*, alternate between 4 and 6. Where the even number exponents end in 6. Therefore* 2024^{2024} *will end in a 6.*

Answer: *6*

10 MARK QUESTIONS

QUESTION 11

What is the first time after 5 o'clock when the angle between the minute hand and the hour hand is 62°?



Answer: *16 minutes past 5 o'clock*

QUESTION 12

Julian and Joseph are twins with different jobs. Julian earns five-eighths of what Joseph earns, but Julian's expenses are half of Josephs's. Julian ends up saving 40% of his income. What percentage of his income does Joseph save?

Let x *represent Julian's income. Then Joseph's income is* $\frac{5}{8}x$*.*

Since Joseph saves 40% of his income, his expenses are 100% — 40% = 60% of his income.

Therefore, Joseph's expenses are $60\% \times \frac{5}{8}x = \frac{60}{100} \times \frac{5}{8}x = \frac{3}{8}x$*.*

Joseph's expenses are one-half of Julian's expenses so Julian's expenses are twice Joseph's expenses.

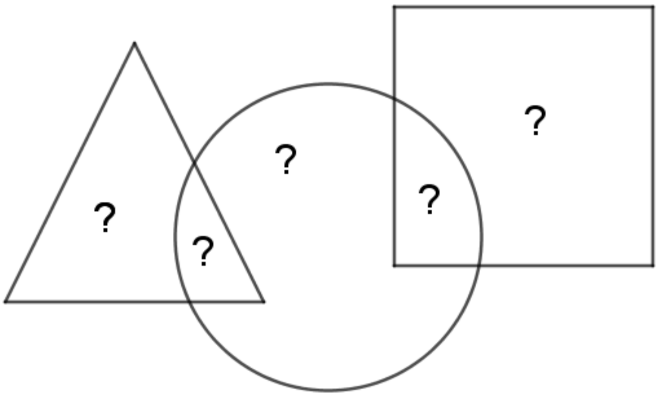
Therefore, Julian's expenses are $2 \times \frac{3}{8}x = \frac{3}{4}x = 0{,}75x = 75\%$ *of* x*.*

Since Julian's expenses are 75% of his income, he saves 100% - 75% = 25% of his income.

Therefore, Julian saves 25% of his income.

Answer: *25%*

QUESTION 13



The question marks represent five consecutive numbers.

When added together:

- The numbers in the triangle = 35
- The numbers in the square = 29

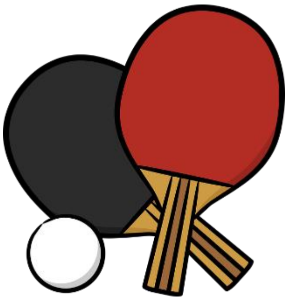
What numbers replace the question marks? *(Order your answer from left to right)*

As the numbers are consecutive, the numbers in the triangle must either be 18 and 17 or 17 and 18. As the numbers in the square only total 29, the numbers must be descending left to right, ie: 18, 17, 16, 15, 14.

Answer:*18,17,16,15,14*

QUESTION 14

Five players compete in a table tennis tournament such that each player plays every other player exactly once. There are no drawers in these matches. In each game both players have the same probability of winning, and the result of one game does not influence the outcome of the others.



What is the probability that some player will win all her games?

$$P(\text{wins all 4 in a row}) = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

$$P(\text{one of 5 wins all 4}) = 5 \times \frac{1}{16}$$

Answer: $\frac{5}{16}$

QUESTION 15

The new 5-rand coin will display the *Southern Right Whale.* If the new R5 coin rolls along the interior of a square along its sides without slipping.

By the time the coin has returned to its starting position, it has made one whole revolution.

What is the side length of the square?

Assume the radius of the coin is 1cm.

The coin starts at a distance of 1 radius, r, from the side of the square. It would then roll a further seven times to get back to its starting position. Therefore the perimeter of the square is the distance the square rolls plus $2r$.

In order to get back to its original starting position it must roll $\frac{1}{4}$ of its perimeter each time it rolls.

$$L = \frac{1}{4} \times 2\pi r + 2r = \frac{\pi}{2} r + 2r$$

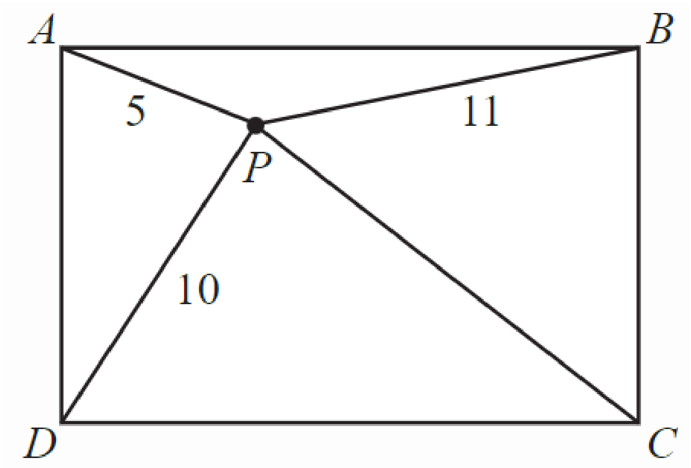
$$= \frac{\pi}{2} + 2$$

Answer: $\frac{\pi}{2} + 2$

20 MARK QUESTIONS

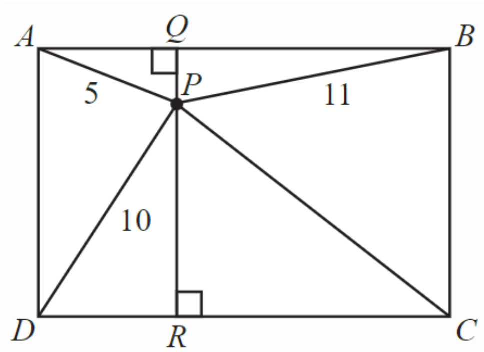
QUESTION 16

Adam, Bonolo, Casey and Delilah are standing on the four corners of a rectangular field, with Adam and Casey at opposite corners. Prish is standing inside the field 5m from Adam; 11m from Bonolo; and 10m from Delilah. In the diagram below, the locations of Adam, Bonolo, Casey, Delilah and Prish are marked with A,B,C,D, and P respectively.



Determine the distance from Prish to Casey.

Start by drawing a line through P, perpendicular to AB and DC. Let Q be the point of intersection on AB and R on DC. We now have $90°$ angles at $A\hat{Q}P; B\hat{Q}P; D\hat{R}P$ and $C\hat{R}P$. We also know that $AQ = DR$ and $BQ = RC$.



$QP^2 = 25 - AQ^2$ *(Pythag)*

And

$QP^2 = 121 - BQ^2$ *(Pythag)*

$\therefore 25 - AQ^2 = 121 - BQ^2$

$\therefore BQ^2 - AQ^2 = 96$

$\therefore CR^2 - DR^2 = 96$

Now: $DR^2 + RP^2 = 100$ *(Pythag)*

$\therefore RP^2 = 100 - DR^2$ *and* $RP^2 = CP^2 - CR^2$

$\therefore 100 - DR^2 = CP^2 - CR^2$ *OR* $CR^2 - DR^2 = CP^2 - 100$

But $CR^2 - DR^2 = 96 \therefore 96 = CP^2 - 100$

$$\therefore CP^2 = 196$$

$$\therefore CP = 14 \ (since\, CP > 0)$$

Answer: *Prish is 14 metres away from Casey*

QUESTION 17

Two trains of equal length are on parallel tracks. One train travels at 40 km/h and the other train travels at 20 km/h.

One day, the two trains are travelling in the same direction, and the front end of the faster train is at the same place as the back end of the slower train.

The faster train then completely passes the slower train so that the back end of the faster train is now at the same place as the front end of the slower train.

Another day, the two trains are travelling in opposite directions, and the front end of the faster train is at the same place as the front end of the slower train. The trains then completely pass each other so that the back end of the faster train is at the same place as the back end of the slower train.

If it takes 2 minutes longer for the trains to completely pass one another when travelling in the same direction than it does when they are travelling in opposite directions, determine the length of each train.

Let L represent the length, in km, of each train.

When travelling in the same direction, the faster train is travelling at 40 − 20 = 20 km/h relative to the slower train.

In order to completely pass the slower train, the faster train must travel $2L$ km. Therefore, it takes $\frac{2L}{20} = \frac{L}{10}$ hours to completely pass the slower train.

When travelling in opposite directions, the faster train is travelling at 40 + 20 = 60 km/h relative to the slower train. In order to completely pass the slower train, the faster train must travel $2L$ km. Therefore, it takes $\frac{2L}{60} = \frac{L}{30}$ hours to completely pass the slower train.

We know it takes 2 minutes longer for the trains to completely pass one another when travelling in the same direction than when travelling in opposite directions.

$$\therefore \frac{L}{10} - \frac{L}{30} = \frac{1}{30}$$

$$3L - L = 1$$

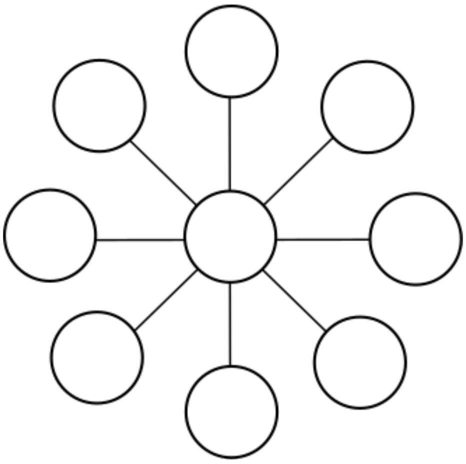
$$2L = 1$$

$$L = \frac{1}{2}$$

Answer: *The length of is ½ kilometre.*

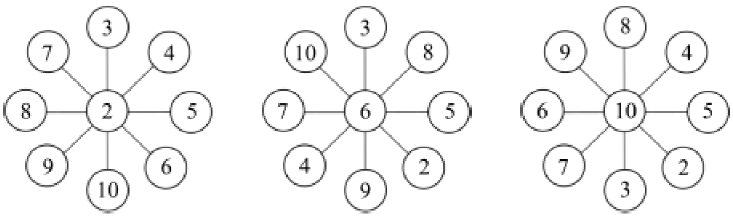
QUESTION 18

Consider the puzzle below in which you are required to place each of the numbers 2, 3, 4, 5, 6, 7, 8, 9, and 10 in a different circle in the diagram so that each line of three circled numbers has the same sum.



Find three numbers that can go in the middle to achieve this.

The three possible numbers to put in the middle are 2, 6 or 10. With middle numbers of 2, 6, or 10, there are many possible ways to place the remaining numbers in the diagram. The only condition is that the pairs of numbers with the same sum must be placed on the same line. Some examples are shown.



If the middle number is 2, the other numbers can be paired as follows: 3 + 10, 4+9, 5+8, and 6+7. Each of these sums is 13, so the sum of any line of three circled numbers would be 13 + 2 = 15.

Similarly, if the middle number is 10, the other numbers can be paired as follows: 2+9, 3+8, 4+7, and 5+6. Each of these sums is 11, so the sum of any line of three circled numbers would be 11 + 10 = 21.

We can also choose 6 as the middle number. Then the other numbers can be paired as follows: 2+ 10, 3+9, 4+8, and 5+7. Each of these sums is 12, so the sum of any line of three circled numbers would be 12 + 6 = 18.

Answer: *Numbers to put in the middle are 2, 6 or 10*

QUESTION 19

Sid and Jim are engaged by the local council to paint yellow lines on either side of a certain street. Sid arrived first and had painted three metres on the right-hand side when Jim arrived and pointed out that Sid should be painting the left-hand side. So Sid started afresh on the left-hand side and Jim continued what Sid had started on the right.

When Jim had finished his side, he went across the street and painted six metres for

Sid, which finished the job. Both Sides of the street were an equal length.

Who painted the greatest distance?

Let L *be the length of the street in metres.*
Jim painted $L-3+6=L+3$ *metres and*
Sid painted $3+L-6=L-3$ *metres.*

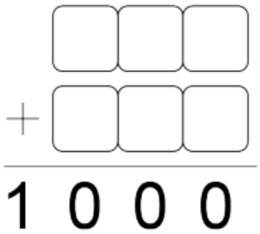
Therefore, Jim painted 6 metres more than Sid. The length of the street is irrelevant.

Answer: *Jim*

QUESTION 20

Ana has nine tiles, each with a different integer from 1 to 9 on it. Ana creates larger numbers by placing tiles side by side. For example, using the tiles 3 and 7, Ana can create the 2-digit number 37 or 73.

Using six of her tiles, Ana forms two 3-digit numbers that add to 1000.



What is the largest possible 3-digit number that she could have used?

To determine the largest possible 3-digit number the top 3 digit number must have a first digit as large as possible, therefore 8 is the only option. The next step is to make the second digit and third digit as large as possible. Therefore 7 and 6.

The bottom number to get 1000 will therefore be 124.

Answer: *876*